



Spaceborne Lidar

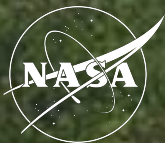
Fundamentals Reference Table

GEDI Data Fundamentals:

[The Global Ecosystem Dynamics Investigation \(GEDI\) mission](#) is a space-based laser altimeter designed to measure vertical structure and topography of Earth's forests with high resolution. GEDI plays a crucial role in advancing our ability to understand atmospheric and water processes, biodiversity, and habitat. Billions of full-waveform observations are collected each year providing 3-dimensional information to forest, weather, water, and fire risk management. Data products for ground elevation, height, canopy cover, plant area index, plant area volume density, foliage height diversity, biomass, and waveform structural complexity [can be found here](#).

References

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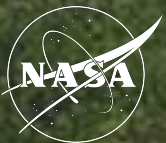




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Variable	Definition	How It Works	Considerations	How It's Formatted
Waveform	The GEDI waveform is the returned energy recorded as a function of receive time. The 'waveform' is the actual captured data, or resulting 'observation' for each 'shot' corresponding to each GEDI 'footprint.'	The active sensor transmits 1064nm near-infrared laser pulse energy from its position on the International Space Station. The energy is reflected and scattered across surface features within the laser footprint as the light pulse travels from the top of the surface feature to the ground. The round-trip time it takes for the energy to reflect off the surface features is converted into distance (range) once properly cleaned and geolocated.	The GEDI lidar system design and processing algorithms are optimized for detecting vegetation vertical structure. The shape of the full waveform is processed with several algorithms to interpret potential distinct layers of the vertical profile. Interpreting full waveforms can be challenging, especially where valid signals are weak, indistinguishable from background noise, or influenced by certain systematic or environmental conditions. Active areas of research continue to improve GEDI's algorithms for resolving details over various surface features.	The L1B product houses the processed and geolocated waveforms as 'rx_waveform'. This is the measured energy for the waveform amplitude as digital numbers (DN) for the relative energy received over a range of height bins. The amplitude values are associated to fixed-size "bins" of time that correspond to the vertical distance/heights. 'Tx' refers to the energy <i>transmitted</i> from GEDI, while 'Rx' refers to the energy <i>received</i> or returned to GEDI.
Shot Number	Unique identifier for each shot/observed waveform at each footprint tagged with associated orbit and beam information.	Each shot receives a number and is available within the relative "Group" of the HDF5 footprint level or other data products.	Retain this identifier when subsetting the footprint level datasets to isolate an individual shot/footprint, and/or explore relationships between variables from different GEDI products.	'shot_number'=OOBBFFNNNNNNNNN: OOOOO: Orbit number BB: Beam number FFF: Minor frame number (0-241) NNNNNNNN: Shot number within orbit 🔴 Pro tip: strictly enforce UInt64 data type for 'shot_number' or cast them as strings when converting data formats.

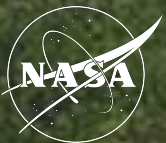




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Beam Type	Refers to two possible beam types built into the sensor. Each type is categorized by the amount of laser beam emitted energy per pulse at mJ/pulse. Higher or full energy (15 mJ/pulse) of the laser generates power beams. Lower energy (4.5 mJ/pulse) emitted by the laser generates coverage beams.	Power: Two independent lasers <i>dither</i> (aka shift position back and forth) to produce two beams each, creating four “full power” beam ground tracks. Coverage: One independent laser is designed to “split” into two coverage beams, reducing the emitted energy per beam. Each of the two beams is also dithered, creating four “coverage” beam ground tracks. This strategy increases spatial coverage, with tradeoffs to resolution, especially over particular features.	Higher energy lasers <i>tend</i> to have a stronger ability to detect the ground and penetrate dense canopies and greater canopy cover. <i>“GEDI was designed to achieve ground detection through 98% canopy cover for strong beams and 95% for coverage beams.”</i> (Dubayah et al., 2020). While the coverage beams were designed to penetrate up to 95% cover, this is only expected under optimal conditions. Users are encouraged to use the power beams in denser vegetated regions or evaluate beam type differences specific to surface, quality, or systematic characteristics.	Each of the eight beams are the higher order group as BEAMXXXX: Power: BEAM0101, BEAM0110, BEAM1000, BEAM1011 Coverage: BEAM0000, BEAM0001, BEAM0010, BEAM0011 For example, if you want to select the data for <u>plant area index</u> from all beams, you would have to select the ‘pai’ variable eight times within the file, following the individual paths in each GEDI02_B file: →BEAM0001 →‘pai’ →BEAM0011 →‘pai’, etc.
Solar Elevation	This parameter indicates the sun’s position at the time of observation.	The system records the solar elevation at the time of observation.	It is recommended to use nighttime observations where possible to reduce the impact of background noise from the sun. Users should evaluate the impacts of day vs. night observations on results over the study area. Removing daytime shots tends to greatly reduce the size of the dataset.	‘solar_elevation < 0’ indicates data collected at night ‘solar_elevation > 0’ is for data collected during the day.





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Algorithm Setting Group (ASG)	The receive waveform interpretation workflow includes a step for applying several waveform interpretation algorithms optimized for processing the waveform over different conditions and surface features. The results are compared and a final "default" ASG is chosen for each footprint.	ASG 1-6 and 10 hold unique combinations of parameters for waveform smoothing width (noise and signal), the waveform signal start and end thresholds, and whether mode filtering was applied. Latitude, longitude, elevation, highest return, detected modes, sensitivity, quality flag, and the metrics themselves have a variation for each ASG calculated.	The ASGs offer several specified interpretations of waveforms given known relationships of those signal returns over particular vegetation types that the user can access and compare across results to optimize GEDI over localized contexts. For example, algorithms with lower signal thresholds improve ground detection by allowing for weaker signals within the waveform interpretation process. This could be beneficial in dense tropical forests. While a low threshold might misinterpret noise as the ground in less dense vegetation covers.	'selected_algorithm' from L2A and up identifies the algorithm used to produce the resulting derived metrics for each shot. Any variable with the suffix '_aN' is identifying the specific algorithm with its respective calculations of that metric or parameter according to the algorithmic results.
Sensitivity	The <i>"maximum canopy cover that can be penetrated considering the Signal to Noise Ratio (SNR) of the waveform"</i> (Hofton & Blair, 2020). Sensitivity is a probability metric "offering insight into the percentage of canopy cover through which we expect to be able to detect the ground 90% of the time" (Duncanson et al., 2020).	An estimate of the relative minimum percentage of the waveform return that needs to be present within the portion of the waveform representing the ground return for the ground to be detected. Waveform sensitivity is calculated by simulating the minimum detectable ground return pulse energy for a given detection algorithm (for each ASG). The area of that minimum detectable ground return is then divided by the total return waveform area resulting in a probability for the waveform ability to detect the ground at that location.	Beam sensitivity is influenced by beam type, and other factors during the time of acquisition such as atmospheric conditions, background solar illumination (day vs. night), and surface reflectance characteristics for the given study area and time period (e.g. topography, canopy cover). Given these influences and their potential confounding behavior, sensitivity can be considered a fundamental indicator of GEDI quality, particularly over areas highly influenced by these factors.	Sensitivity will have a range of values, but for the waveform interpretation algorithm to run, sensitivity will need to be > 0 and < 1. Alternative calculations of sensitivity for each ASG are found as 'sensitivity_aN'.

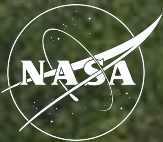




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Quality Flag	A binary flag for users to remove or lower outlier or low quality returns.	A value of 1 indicates the laser shot meets criteria for energy, sensitivity, amplitude, and real-time surface tracking quality and distinguishability for the algorithm to run.	Each data product either attaches the quality flag metrics or applies the quality filtering to the datasets to meet recommended requirements for systematic, and/or land cover criteria, depending on the dataset.	'quality_flag' is referred to as: 'l2a_quality_flag' in higher products, L2B: 'l2b_quality_flag', L4A: 'l4_quality_flag', 'wsci_quality_flag', etc. Alternative calculations of the quality flag for each ASG include the '_aN' suffix.
Elevation Highest Return	The estimated elevation of the highest detected return of the receive waveform. This metric is used to represent the canopy top position.	The geolocated waveforms undergo processing to distinguish signal from noise. The highest and lowest detected returns become the boundaries where modes and other waveform characteristics between these points are processed in order to later calculate higher level metrics like canopy height.	The ASGs can contribute to different interpretations of the highest return. The accuracy of the measured elevation highest return can also be impacted by conditions like low-lying clouds being mistaken for canopy, errors in ground detection, or challenging interpretation over complex spatial variation in canopy heights.	'elev_highestreturn' Alternative calculations of this variable for each ASG are found as 'elev_highestreturn_aN'.
Elevation Lowest Mode	The estimated elevation of the center of the lowest identified mode in the received waveform. This mode represents ground elevation.	Elevation lowest mode is the center of the mode the processing algorithm identifies with peak energy distribution over the ground between the highest and lowest returns.	The ASGs can contribute to different interpretations of the detected lowest mode of the waveform. The accuracy of the elevation lowest mode of can also be impacted by conditions affecting waveform characteristics like cloud interference, mixed or weak ground returns to other signals like low canopies.	'elev_lowestmode' Alternative calculations of this variable for each ASG are found as 'elev_lowestmode_aN'.





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Latitude & Longitude of the Lowest Mode	The final latitude and longitude located for the lowest mode of the waveform.	Once the lowest mode has been interpreted, its latitude and longitude are linearly interpolated between the L1B geolocated waveform first and last sample bins (ranging points).	The geolocation accuracy is ~ 10.2m. Geolocation can be improved by matching the waveforms to simulated GED1 waveforms (require airborne lidar), or referencing the elevation lowest mode to high-quality Digital Elevation Models to calibrate the data. However, if no airborne lidar or reliable DEMs overlap with the data, alternative measures for assessing and correcting for these areas may need to be deployed by the user.	'lat_lowestmode' and 'lon_lowestmode' Alternative calculations of this variable for each ASG are found as: 'lat_lowestmode_aN', and 'lon_lowestmode_aN.'
Number of Detected Modes	Quantifies the number of distinct surfaces features reflected within the waveform within the footprint location.	The conditions from when the waveform was collected, how the waveform was smoothed/denoised, interpreted by each ASG and mode filtering applied or not, are all factors determining the number of modes detected in the waveform processing.	Multi-modal waveforms are a strong indication of complex surface characteristics such as what is found in vegetation or irregular topography. Mode detection can be influenced by the strength of the SNR at both ground and canopy (associated with power and coverage beam tendencies). Cloudy conditions and solar elevation can also impact the number of modes captured. Dense canopies and high slopes can potentially over complicate the waveform shape and confuse distinguishing characteristics of the modes across the vertical profile.	'num_detectedmodes' Filter the datasets with 'num_detectedmodes = 0' to remove "empty" waveforms. Alternative calculations of this variable for each ASG are found as 'num_detectedmodes_aN.' Exploring the results from different ASGs may help reduce inclusion of noisy.





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Degrade Flag	Tagged observations where degradation of the pointing or positioning of the system/sensor may have impacted the horizontal geolocation accuracy.	The degrade status conditions specific to systematic or observational conditions are tracked by the system.	Degraded conditions can lead to excessive horizontal geolocation accuracy . Some studies have found a more detailed selection for removing observations over specific degraded conditions that were particularly more impactful, i.e. filtering out specific non-zero degrade statuses but keeping others, helped retain valid shots rather than assuming all degraded shots hold invalid data by filtering as > 0 .	' degrade_flag = 0 ' indicates the shot is not degraded. Shots with non-zero flags include a range of specific tracking attributes. Non-zero tens digits indicate degraded attitude , and non-zero units digits indicate degraded trajectory .
Local Beam Elevation	This is the GEDI viewing angle (θ) at which the GEDI laser pulse is received relative to nadir. The angle is from the local horizontal plane upwards toward GEDI from the ground bounce point.	For GEDI, the zenith-viewing angle (off-nadir angle) is kept at 6 degrees or less for accurate ranging . As a result, the local beam elevation typically lies between 84-90 degrees .	Consistent angles are expected to have little to no impact on accuracy of the datasets. Areas of highly variable topography or elevation changes (e.g. water surfaces) can potentially have variable accuracies. Some studies have found, though, that filtering datasets for local_beam_elevation ≥ 1.5 radians (keeping high angles) can be beneficial in scenarios where ground elevation accuracy may be impacted (Fayad et al., 2020; Tommasso et al., 2023).	' local_beam_elevation ' The viewing angle can differ from one beam to another on the given date.
Leaf Status Flag	Leaf status is based on ancillary data indicating the phenological periods classified as leaf-on, leaf-off, or transitional stages during observation.	The classification is based on relative greenness derived from normalized NDVI values, and Julian days (for deciduous forests) .	The presence or absence of leaves inherently affects the lidar signal characteristics and ability to capture complexities in GEDI's structural metrics, possibly leading to systematic underestimation of these metrics during leaf-off conditions. Non-deciduous strata such as evergreens, woodlands, grasslands, shrublands, etc. by contrast do not exhibit the same limitations across seasons when considering leaf presence.	' land_cover_data/leaf_off_flag = 0 ' indicates leaf-on, ' land_cover_data/leaf_off_flag = 1 ' indicates leaf-off status.

